



GROW_SCI_A2_W01 · Why day turns to night · teacher note

Objective (verbatim): Explain day and night and the apparent movement of the Sun.

Scheme of work: GROW scheme of work, Autumn 2 week 1.

Reading band: GROW 8–11. Rough whole pupil-visible ARI grade 2.3; age guide 8.3 years, 842 words (guide text excluded). Screening estimate only. Count pupil-visible text excluding data-mbm-guide; science terms are pre-taught.

Structure: Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3.
One-minute resets within Arrival, We do, Independent.

Retrieval and success

The first Autumn 2 week retrieves three cells from the Autumn 1 SoW sequence. “Last term” in the pupil prompt means the preceding Autumn 1 teaching block; check the class sequence and change that prompt if the sequence was not taught.

Time recalled	Question	Expected answer	SoW cell
last lesson	What goes around Earth?	The Moon.	GROW scheme of work, Autumn 1 week 7
2-4 weeks	Name one mechanism from our forces work.	A lever, pulley or gear.	GROW scheme of work, Autumn 1 week 4
last term	Name one force from our first science week.	Gravity, air resistance or friction.	GROW scheme of work, Autumn 1 week 1

Success criteria shown at Starter and checked at Exit: I can point to the day and night sides. · I can turn the model and explain when night comes. · I can explain why the Sun seems to move.

Show-me check after I do: Show me: where is the dot when that place has day? Answer: On the side facing the torch or light.

Words to pre-teach

Definitions and examples match both pupil files.

Word	Tier	Meaning	Example
rotation	Tier 3	one turn around an axis	Earth’s rotation brings day and night.
axis	Tier 3	an imagined line a planet spins around	Earth turns around its axis.
shadow	Tier 2	an area where an object blocks light	The ball makes a shadow on the desk.
orbit	Tier 3	a curved path around another object	Earth’s orbit takes it around the Sun.

Model, challenge and stretch

I do full model: Torch still → dot faces the torch: day → turn the ball → dot faces away: night. Earth’s rotation makes the Sun seem to move.



We do part-done: Torch still → dot faces the torch: day → turn the ball → dot faces away: .

Removed step: Name night when the dot faces away from the torch.

Fade next time: Next time omit the day label too; ask pupils to name both positions before word cards.

Independent has no worked steps; adults can help handle kit without supplying the missing reasoning.

We do F support route: Point to the torch and dot before naming the missing step.

We do M route: Fill the blank and explain why.

We do H stretch route: Explain why the torch need not move to change the view at the dot.

Tier	Task	Speaking stem
F	Point to the lit side and the dark side.	I point to _ for day and _ for night.
M	Explain what the dot sees as Earth turns.	The dot has night when _ because _.
H	Explain why the Sun seems to move across our sky while the model torch stays still.	The Sun seems to move because _. The model shows _.

Go deeper: Mark two dots on opposite sides. Explain why one has day while the other has night.

Named early-finisher stretch — Two dots, one turn: Two dots, one turn: test the claim that the Sun goes round Earth each day. Show the motion that explains both dots. Name one limit of the torch model.

Hinge at We do 2

Ask: On a new ball model, the red dot has just turned from light into shade. What does the shade represent? All pupils respond together by whiteboard, finger-vote, symbol card, thumbs or point. Do not auto-grade or save.

Option	Correct?	Diagnosis
Day at the red dot.	no	Keeps the old day label after the dot has rotated into shade.
The Sun turning off now.	no	Treats darkness at one place as the Sun switching off.
The whole Earth going dark.	no	Extends one dot's night to every place on Earth.
Night at the red dot.	yes	Uses the new case accurately.

Decision: About 3 in 4 or more right → move on. Otherwise re-model with the dot on the far side of the ball, then ask the parallel hinge.

Parallel hinge: A blue dot turns from shade into torchlight. Which part of a day does this represent? Answer: daytime at that dot.

Two likely misconceptions:

Misconception	Spot	Correction
Night at one dot means night everywhere on Earth.	Pupil chooses C, "The whole Earth going dark."	Turn the ball: one dot can be in shade while another still faces the light.



Misconception	Spot	Correction
The Sun switches off when our place has night.	Pupil chooses B, "The Sun turning off now."	Keep the torch on and turn the ball; the dot moves into shade.

Planned adaptations

EAL is language access, not SEN. Keep the same M/H thinking.

Need area	Concrete adaptation	EEF five-a-day strategy
Cognition & Learning	Use a three-step dot and torch card. Omit its last word, then fade the card.	scaffolding
Communication & Interaction	Point to the model before a partner rehearsal with the M or H speaking frame.	scaffolding
SEMH	Offer a known kit role, a quiet pause cue and one small re-entry step.	flexible grouping
Sensory/Physical	Use a stable stand or a large tactile dot; a photo can capture a genuine demonstration.	technology
EAL	Pre-teach science words with the diagram. Plan in any language; use M or H speaking frame before writing. EAL is language access, not a reason to move down to F.	explicit instruction

Talk before write: The dot has night when _ because _ . / The Sun seems to move because _ ; the model shows _ . Plan in any language. Do not machine-translate.

Practical and safety

Prediction: Predict: if the dot turns away from the torch, will it have day or night?

Kit: Torch, Ball or globe, Removable dot, Stable stand if needed

Method:

1. Place the dot on the ball.
2. Keep the torch still and aim it at the ball, away from eyes.
3. Turn the dot to face the torch; record day or night.
4. Turn the dot away; record day or night.

Results table: dot faces torch / day or night; dot faces away / day or night.

Risk: Check against the school risk assessment and CLEAPSS guidance. Do not shine the torch into eyes.

This is a classroom model used in this lesson. The SoW row does not expressly require a practical.

Entry Level style bridge: explain a change seen in the model. This is a planning route, not an official qualification outcome or unit code.

Pathway expectations

Pathway	Observable response in this routine
Pathways	High staff support; VF and WS seen often; begins to respond to one simple next step.



Pathway	Observable response in this routine
Build	A mix of WS and I; can say or show what feedback meant.
Grow	Mostly independent; self-edits at // and responds to NS(+) with R.
Launch	Independent, self-assesses against the three success criteria.

Adults, feedback and handoff

At lesson start choose teacher or TA for the supported pupils during Independent. The other adult holds the group. Swap across the week.

Stage	TA Focus line
I do	Watch the dot. Point to it when it faces the light.
We do	What is done already? What step is left for you?
Independent	Show your first step. What can you do on your own next?

Prompt ladder without touch: Wait and count to 5. → Point to the dot or light. → Ask: What changes at the dot? → Model one turn; let the pupil try. → Finish one example together, then return the next turn to the pupil.

Move back to pupil action as soon as possible. At Exit hand over the highest prompt level, what was independent, and whether the next step was acted on.

Cold-call bank (use the agreed no-public-pressure route where needed):

Tier	Question	Answer
F	Point to day on the model.	The lit side.
F	Point to night.	The side away from light.
M	What moves in our model?	The ball turns.
M	Why does the dot get night?	It turns away from the light.
H	Why does the Sun seem to cross the sky?	Earth turns, changing our view.
H	What does the torch model leave out?	It simplifies size, distance and Earth's orbit.

Codes: VF · WS · I · NS(+) · E · R · // · ?; E: evidence captured on the school's evidence app only when real evidence occurs.

VF for verbal feedback; WS for supported work; I for independent work; NS(+) at Exit; // for a Review self-edit; ? for a read-back; R when the pupil acts on the next step; E only if actual evidence was captured. Record the hinge and one precise next step in the staff lesson log. Never collect evidence to create distress.

Regulation: Agree a quiet pause signal and supervised place. Check in before the hinge and Independent; offer one re-entry step. No evidence while a pupil is distressed.

QR poster: GROW_SCI_A2_W01 (lesson record; staff device or projector only)

Gatsby 4: Weather observer: know which places have daylight when checking observations.



Assistive technology: speech-to-text, text-to-speech or immersive reader; response may be point, sign, speech, draw, type, write, scribe or a staff-captured photo of genuine work.

The teacher or TA hears the pupil at Exit and changes the next example or support. Ask: What are you learning? What feedback did you get? What will you improve? How do you know it worked?

Source review

Every final science claim in the pupil page, organiser and SVG labels is represented in LESSON_SOURCE.json facts, with named source and URL. The diagram and torch model are original constructed teaching material. No local claim is made.

Pupil-facing risk: Torch beams can dazzle eyes; a loose ball may roll away.

Pupil-facing safety: Aim the torch at the ball, never at eyes. Keep the ball on the tray or stand.

SLT planning and evaluation

Lesson code: SCI-GROW-A2-W01

SoW reference: GROW scheme of work, Autumn 2 week 1

Exact SoW outcome: Explain day and night and the apparent movement of the Sun.

Evidence (scheme of work): Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W01. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Place a dot on the ball. Point to the side lit by the torch. Staff: Offer a seat or standing spot. End with a one-minute reset.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes questions without a public test.

I Do (5 min): Watch: the torch stays still. I turn the ball. The dot has day when it faces the light. It has night when it faces away. Staff: Model the full sequence. Keep light out of eyes. Ask the show-me check.

We Do (7 min): Now try the same model. Two steps are done. Complete the last step: turn the dot away and name what comes next. Staff: Wait for the missing step. End with a one-minute reset.

We Do 2 (5 min): Turn the ball. Choose where the dot is when it is night. Show your choice with the group. Staff: Use an all-pupil hinge response and the decision rule in the note.

Independent (8 min): Show day and night at the dot. Explain why the Sun seems to move, even when our torch stays still. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset at end.

Review (4 min): Tell a partner what your model shows. Hear one comment. Change one part and say what you changed. Staff: Give one specific comment, invite a self-edit (/) or read-back (?).

Exit (3 min): Show how you know. Tell us your feedback and next step. Say, point, sign, draw or write. Staff: Hear answer, choose next model or help; hand over highest prompt and independent part.



Planned adaptation — cognition learning: Use a three-step dot and torch card. Omit its last word, then fade the card.

Planned adaptation — communication interaction: Point to the model before a partner rehearsal with the M or H speaking frame.

Planned adaptation — semh: Offer a known kit role, a quiet pause cue and one small re-entry step.

Planned adaptation — sensory physical: Use a stable stand or a large tactile dot; a photo can capture a genuine demonstration.

Planned adaptation — eal: Pre-teach science words with the diagram. Plan in any language; use M or H speaking frame before writing. EAL is language access, not a reason to move down to F.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: scaffolding; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Show day and night at the dot. Explain why the Sun seems to move, even when our torch stays still. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W02. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W02: Describe the Sun, Earth and Moon as approximately spherical bodies.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Name night when the dot faces away from the torch. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "On a new ball model, the red dot has just turned from light into shade. What does the shade represent?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Place a dot on the ball. Point to the side lit by the torch. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Show day and night at the dot. Explain why the Sun seems to move, even when our torch stays still. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use rotation, axis, shadow, orbit with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.



Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.